

STEM Reasoning Lab

Raw synthetic question source — no answer key

Q1 — Calculus

5 points

Evaluate

$$\int_0^1 x e^{x^2} \, dx.$$

Show the substitution and the transformed bounds.

Q2 — Mechanics

8 points

A 2 kg block slides 3 m from rest down a 30° incline with $\mu_k = 0.20$. Find its acceleration and final speed. Use $g = 9.81 \, \text{m s}^{-2}$.

STEM Reasoning Lab

Raw synthetic question source — no answer key

Q3 — Linear algebra

7 points

For any real $m \times n$ matrix A , prove

$$\ker(A) = \ker(A^T A),$$

then conclude $\text{rank}(A) = \text{rank}(A^T A)$. Do not assume that A is square or invertible.

STEM Reasoning Lab

Raw synthetic question source — no answer key

Q4 — Programming

10 points

Implement `stable_softmax(xs)`. Return `[]` for empty input and avoid overflow for values near 1000. Do not use NumPy.

Required signature

```
def stable_softmax(xs):  
    ...
```